Session 6: Solutions and facilitation notes

Foundations of Clinical Research · instructor copy (do not distribute before the session)

Exercise numbers follow `Practicals/s6_exercises.md`. All FLUID-ICU numbers come from the results packs in `Course/Data/results_packs/` (simulated data). Each answer names the pack table it comes from.

Exercise 1: True or false

#	Answer	One-line explanation
1	**T**	That is the practical reading of a 95% CI: the range of values compatible with the data.
2	**F**	Absence of evidence is not evidence of absence. The CI (−15.2 to 41.6 ml/kg) is wide: the pilot cannot tell.
3	**F**	p is calculated *assuming* the null is true. It is the probability of data at least this extreme under the null, not the probability of the null.
4	**T**	A 95% CI that includes the null value corresponds to $p > 0.05$ (two-sided), and vice versa.
5	**T**	More patients give more precision, and so a narrower CI.
6	**F**	A tiny p can accompany a trivial effect in a large study. Judge size from the estimate and CI.
7	**T**	This is the strict definition: the 95% refers to the method over repeated studies (the "dance of the CIs").
8	**T**	A small study has low power, so a real effect can be missed (missed fire).
9	**T**	Alpha = type I error rate = risk of a false alarm.
10	**F**	p-values do not "trend". Report the estimate and CI and call it what it is: not statistically significant.
11	**T**	With 20 independent tests and no real effect, the chance of at least one $p < 0.05$ is about 64% (the jelly-bean problem).
12	**F**	Same patients measured twice: use a **paired** test (paired t-test, or Wilcoxon signed-rank if skewed).

#	Answer	One-line explanation
13	**F**	Switching to one-sided after seeing the data halves the p-value and is not acceptable. Two-sided is the default. A one-sided test needs a strong reason, written in the protocol before the data.
14	**F**	Non-inferiority is a different design: a margin fixed in advance, its own hypotheses, sample size and analysis (Session 3).

Exercise 2: Choose the test

#	Test	Why?
a	**Independent (unpaired) t-test**	Numerical, roughly symmetric outcome, two independent groups.
b	**Wilcoxon rank-sum (Mann-Whitney U) test**	Numerical but skewed, small groups. (Same choice as Paper 1 for fluid balance.)
c	**Chi-square test**	Binary outcome, two independent groups, large expected counts.
d	**Fisher's exact test**	Binary outcome, very small counts (expected counts < 5).
e	**Paired t-test**	Same patients measured twice, and the differences are roughly symmetric. (Wilcoxon signed-rank if skewed.)
f	**McNemar test**	Binary outcome measured twice in the same patients (paired categorical).

Instructor note: if someone suggests "chi-square" for (f), ask "are these two independent groups?" They are not. The same patients are measured twice, so the data are paired and McNemar applies.

Exercise 3: Match r to the scatter plots

Plot	r
A	**−0.58**: points fall from left to right, moderate scatter
B	**0.01**: round cloud, no straight-line trend
C	**0.86**: tight rising band
D	**0.30**: weak rising trend, lots of scatter

Bonus: **no**, not on correlation alone. High r means the two methods rise and fall together, not that they give the same value. Look at **agreement**: a Bland–Altman plot (difference against average), the mean difference (bias) and the 95% limits of agreement. Then decide whether those differences are clinically acceptable.

Instructor note: the plotted data were simulated with exactly these correlations.

Exercise 4: [FLUID-ICU] Interpret the comparison

In class, everyone fills in the table for Q1 alone. Participants fill it in again for their chosen question at home.

The table for Q1 (Q1 results pack)

	Model answer
Outcome and its type	Death from any cause by day 28 (`died_28d`), binary
Test, and why this test	Chi-square test: a binary outcome in two independent groups with large counts. (The pack does not name the test.)
Result in the negative-balance group	45/245 (18.4%) died (Table 2)
Result in the positive-balance group	134/373 (35.9%) died (Table 2)
Effect with 95% CI	Risk difference 17.6 percentage points (95% CI 10.7 to 24.4) (Table 2)
p-value	< 0.001 (Table 2, chi-square p)
One reason this unadjusted comparison could be biased	Confounding by indication. In Table 1 the positive-balance group was sicker: median SOFA 8 (6, 10) vs 6 (4, 8), vasopressor 205 (55%) vs 61 (25%). Sicker patients receive more fluid and also die more.

Model Results sentence (Q1): "By day 28, 134 of 373 patients (35.9%) with a positive fluid balance and 45 of 245 (18.4%) with a negative balance had died (risk difference 17.6 percentage points, 95% CI 10.7 to 24.4, p < 0.001)."

Instructor note: Table 2 also gives RR 1.96, without a CI. The crude logistic regression in Table 3 gives the same p < 0.001.

When participants redo the table for their chosen question (Q2 to Q5)

Q	Outcome and type. Test	Negative-balance group	Positive-balance group	Effect with 95% CI	p-value
Q2	AKI, any stage (`aki_any`), binary. Chi-square test (the pack does not name it)	71/245 (29.0%) (section 4)	230/373 (61.7%) (section 4)	Crude OR 3.94 (2.80 to 5.60) (Table 4)	< 0.001 (Table 4)
Q3	Ventilator-free days to day 28, a count piled up at 0 and 28. Wilcoxon rank-sum (Mann-Whitney U) test	Median 28 (22 to 28) days (Table 2)	Median 20 (0 to 28) days (Table 2)	Hodges–Lehmann difference −3.0 days (−5.0 to 0.0) (Table 3)	< 0.001 (Table 3)

Q	Outcome and type. Test	Negative-balance group	Positive-balance group	Effect with 95% CI	p-value
Q4	Death by day 28, binary. Lactate is a number. Unadjusted logistic regression (Table 4)	Survivors: median lactate 2.3 (1.6, 3.4) mmol/L (Table 1)	Died: median lactate 3.1 (2.2, 4.5) mmol/L (Table 1)	OR per 1 mmol/L 1.38 (1.23 to 1.55), complete case, n = 527 (Table 4)	< 0.001 (Table 4)
Q5	ICU length of stay in survivors, right-skewed. Wilcoxon rank-sum (Mann-Whitney U) test	Median 4.8 (3.3 to 7.1) days (Table 2)	Median 7.5 (5.2 to 11.2) days (Table 2)	Hodges–Lehmann difference 2.5 days (1.8 to 3.2) (Table 3)	< 0.001 (Table 3)

One reason for bias, from Table 1 of each pack:

- **Q2, Q3 and Q5:** confounding by indication. The positive-balance group was sicker at baseline (median SOFA 8 vs 6).
- **Q4:** patients who died were older and sicker (mean age 66.0 vs 61.2 years, median SOFA 8 vs 7). Lactate may partly stand for general severity.

Model Results sentences:

- **Q2:** "AKI occurred in 230 of 373 patients (61.7%) with a positive balance and 71 of 245 (29.0%) with a negative balance (crude OR 3.94, 95% CI 2.80 to 5.60)."
- **Q3:** "Patients with a positive balance had fewer ventilator-free days (median 20 vs 28 days, Hodges–Lehmann difference −3.0 days, 95% CI −5.0 to 0.0)."
- **Q4:** "Median admission lactate was 3.1 mmol/L in patients who died and 2.3 mmol/L in survivors (unadjusted OR 1.38 per 1 mmol/L, 95% CI 1.23 to 1.55)."
- **Q5:** "Among survivors, the median ICU stay was 7.5 days with a positive balance and 4.8 days with a negative balance (Hodges–Lehmann difference 2.5 days, 95% CI 1.8 to 3.2)."

Exercise 5: Read a regression table (another example, homework)

1. **Binary outcome** (delirium yes/no), so use logistic regression, which reports odds ratios.
2. **1** (a ratio). An OR of 1 means equal odds.
3. CI excludes 1 (□) or includes 1 (□):

Row	OR (95% CI)	Excludes 1?
Benzodiazepine	2.10 (1.40–3.20)	□
Age per 10 y	1.30 (1.12–1.51)	□
Mechanical ventilation	1.60 (0.95–2.70)	□
Female sex	0.85 (0.60–1.20)	□
Dementia	3.40 (1.90–6.10)	□
Sleep protocol	0.62 (0.41–0.94)	□

4. Model sentence: *"After adjusting for age, sex, ventilation, dementia, sleep protocol and illness severity, patients who received a benzodiazepine infusion had about twice the odds of developing delirium (OR 2.10, 95% CI 1.40 to 3.20)."*

5. The OR multiplies for each 10 years: 20 years older gives 1.30 × 1.30 ≈ **1.69 times the odds** (all else equal).

6. **Mechanical ventilation** and **female sex** (CIs include 1).

7. Problems in the conclusion:

- **Causal language** ("cause", "reduces") from an observational cohort: it should be "associated with". Confounding by indication is likely, because sicker or more agitated patients receive benzodiazepines.
- **"Trend"** for p = 0.08: the CI (0.95–2.70) includes 1, so it is not statistically significant. Report the estimate and CI.
- **"Reduces delirium by 38%"** confuses odds with risk. OR 0.62 is a 38% lower *odds*, not risk. Delirium is common, so the risk reduction is smaller. It also implies causation, and the sleep protocol was not randomised.

Instructor note: the table is mock data. The p-values are consistent with the CIs (computed from the log-scale standard error).

Exercise 6: Read a Kaplan–Meier curve

1. About **82%** alive in the restrictive group and about **67%** in the usual-care group at day 28 (read where each curve ends). Accept ±3%.

2. Each tick is a **censored** patient: follow-up stopped (lost to follow-up, transferred, withdrew) without death being observed. They count as alive up to that day.

3. **67** usual-care patients were at risk at day 21. The number falls because of **deaths and censoring**. In the restrictive group 16 died but 33 left the risk set (from 100 to 67 at day 28), because 17 were censored before day 28.

4. The log-rank test gives evidence that the two survival curves differ (p = 0.009: results this different would be unusual if the strategies had the same survival). It does **not** give the size of the effect, its 95% CI, or any adjustment for confounders.

5. **Cox regression** gives the hazard ratio with a 95% CI. Check the **proportional-hazards** assumption: curves that cross or converge strongly suggest the HR is not constant over time.

6. No. Without numbers at risk the reader cannot see how many patients support each part of the curve. The right-hand end may rest on very few patients. Reporting guidelines and reviewers expect them.

Instructor note: the trial is simulated, and the values were computed from the plotted data.

Exercise 7: Why is it missing? MCAR, MAR or MNAR

#	Mechanism	Consequence for complete-case analysis
a	**MCAR**: an accident unrelated to the patients	Unbiased, just less precise (fewer patients)

#	Mechanism	Consequence for complete-case analysis
b	**MAR**: depends on admission time, which is recorded	Possibly biased if night admissions differ. Multiple imputation using admission time can correct it.
c	**MNAR**: depends on the missing value itself (the worst pain)	Biased (pain looks better than it is). No guaranteed fix: run a sensitivity analysis.
d	**MNAR** (deteriorating patients are transferred). At best MAR, if the reason for transfer is recorded.	Biased (mortality underestimated). Chase day-28 status by phone or records, and run a sensitivity analysis.
e	**MCAR** (missing by design for one month, unrelated to patients), unless patients in the first month differed	Usually unbiased. Check whether early patients differed.
f	**MAR**: depends on age, recorded for all	Biased towards younger patients' quality of life. Use multiple imputation including age.

Instructor note: for (d), a sensitivity analysis can report the main result plus a best case (all transferred patients alive) and a worst case (all died). If the conclusion holds in both, it is robust.

Exercise 8: [FLUID-ICU] Interpret the model

In class, everyone does Part A and the two sentences for Q1. Parts B and C are homework for Q1.

Part A: crude vs adjusted (Q1, Table 6 of the Q1 pack)

	Estimate	95% CI	n
Crude	OR 2.49	1.70 to 3.70	618
Adjusted	OR 1.62	1.02 to 2.59	527

1. Age, SOFA score, lactate, vasopressor, mechanical ventilation and chronic kidney disease. Yes, they were chosen before the analysis: section 2 lists them as "Pre-specified confounders (SAP)".

2. The OR fell from 2.49 to 1.62, towards 1 (no effect). In the same 527 patients the crude OR was 2.22 (Table 6), so the fall comes from the adjustment, not from the smaller sample. The positive-balance group was sicker, so part of the crude association was confounding by indication.

3. No. The adjusted CI (1.02 to 2.59) excludes 1, but only just. The association is modest and imprecise.

Part B: complete case vs multiple imputation (Q1, Table 6)

4. 527 patients in the complete-case model and 620 in the multiple-imputation model. The complete-case model left out the 91 patients without a lactate (or age) value, and the 2 without a day-3 fluid balance (participant flow, section 2).

5. Yes. Multiple imputation gave OR 1.70 (95% CI 1.09 to 2.64), close to the complete-case 1.62 (1.02 to 2.59).

Part C: Kaplan–Meier and Cox (Q1)

6. Survival at day 28 was 81.6% with a negative and 64.1% with a positive balance, log-rank p < 0.001 (Table 2). The adjusted Cox HR was 1.46 (95% CI 1.00 to 2.13), p = 0.053 (Table 4). The logistic and Cox results tell the same story: same direction, similar size and overlapping CIs. Whether p falls just below or just above 0.05 does not change the interpretation.

Model Results sentences (Q1): "Positive fluid balance was associated with higher 28-day mortality (crude OR 2.49, 95% CI 1.70 to 3.70). After adjustment for age, SOFA, lactate, vasopressor use, mechanical ventilation and chronic kidney disease, the association was smaller (adjusted OR 1.62, 95% CI 1.02 to 2.59)."

When participants redo Part A for their chosen question (Q2 to Q5)

Q	Crude estimate (95% CI), n	Adjusted estimate (95% CI), n	Source
Q2	OR per 10 ml/kg 1.37 (1.27 to 1.47), n = 618	1.27 (1.16 to 1.40), n = 527	Table 4
Q3	Difference per 10 ml/kg −1.48 (−1.84 to −1.13) days, n = 618	−0.61 (−1.00 to −0.21) days, n = 527	Table 3
Q4	OR per 1 mmol/L 1.38 (1.23 to 1.55), n = 527	1.39 (1.24 to 1.57), n = 527	Table 4, complete case
Q5	Ratio of geometric means 1.50 (1.34 to 1.68), n = 439	1.24 (1.10 to 1.39), n = 366	Table 3

- **Question 1.** Q2, Q3 and Q5 adjust for the six confounders listed in section 2 as pre-specified in the SAP. Q4 adjusts for age, sex, diabetes, chronic kidney disease and heart failure (Table 4).
- **Question 2.** In Q2, Q3 and Q5 the adjusted estimate is closer to no effect than the crude one, so part of the crude association was confounding by indication. In Q4 the estimate hardly moved (1.38 to 1.39), because the main model does not include severity. Adding SOFA brings it to 1.21 (Table 4).
- **Question 3.** No. All four adjusted CIs exclude the no-effect value (1 for Q2, Q4 and Q5, 0 for Q3).
- **Part B for Q4 (Tables 2 and 4).** 527 patients in the complete-case model and 620 with multiple imputation. The 93 patients without a lactate were left out. They were less sick (median SOFA 5 vs 8) and died less often (18 (19%) vs 161 (31%)). Multiple imputation gave OR 1.45 (1.29 to 1.64), close to the complete-case 1.39 (1.24 to 1.57).

Exercise 9: Proposal builder 6 (what a good answer looks like)

The instructor fills this in for the running example, a trial of a restrictive fluid strategy against usual care.

9.1 Sample-size statement

Input	Value	Source
Expected value in the comparison group	40% die by day 28 with usual care	Unit audit or published septic-shock trials
Smallest difference worth detecting	10 percentage points (40% to 30%)	Clinical judgement: the smallest fall that would change practice

Input	Value	Source
SD (only for a numerical outcome)	Not needed: the outcome is binary	
Alpha (two-sided)	5%	convention
Power	80%	convention
Expected drop-out	10%	Pilot or audit of loss to follow-up

> Assuming **40%** mortality in the control group and **30%** in the intervention group, **356** patients per group give **80%** power at a two-sided alpha of 5%. Allowing **10%** loss, we will recruit **792** patients in total.

(356 per group from the spreadsheet, sheet 2. Dividing by 0.90 gives 396 per group, 792 in total.)

9.2 Plain-language analysis plan

Step	What we will do
Describe the patients (Table 1)	Numerical symmetric variables as mean (SD), skewed as median (IQR), categories as n (%)
Compare the primary outcome	Test: chi-square test because the outcome (death by day 28) is binary
Report the effect	Risk ratio and risk difference, each with its 95% CI
Deal with confounding (observational designs only)	Not needed in this randomised trial. In an observational design: adjust for age, SOFA and lactate using logistic regression
Missing data	Report the number missing for each variable. Prevent it by a short CRF with a "not done" box, weekly checks of missing items, and a phone call to find day-28 status

9.3 Confounders

In a randomised trial, randomisation balances the confounders. For an observational version of the same question:

Confounder	Why it is a confounder
Age	Older patients often receive less fluid, and they die more. Age is not caused by fluid.
SOFA score at admission	Sicker patients receive more fluid and die more. It is measured before the fluid.
Lactate at admission	A marker of shock: it drives fluid giving and predicts death. It is measured before the fluid.

Expected number of events: about 277 deaths (40% of 396 plus 30% of 396). Maximum number of model terms: about 27.

9.4 Missing-data plan

Question	Our answer
Which variables are most likely to be missing, and why?	Lactate at admission, which is less often measured in less sick patients (MAR, explained by SOFA). Day-28 status in patients transferred to another hospital (possibly MNAR).

Question	Our answer
How will we prevent it?	Lactate in the admission bundle, a "not done" box on the CRF, weekly checks of missing items, and a phone call at day 28
How will we report missingness?	n missing per variable in Table 1, and a flow diagram showing losses

Instructor note: quick check when reading drafts.

Element	Good enough	Needs work
Sample size	4 inputs with a stated source	"All patients in one year"
Analysis plan	Summary + test/model + effect measure with 95% CI	"Data will be analysed with SPSS"
Confounders	Named, justified, within events-per-variable	Stepwise selection, mediators adjusted
Missing data	Expected gaps + prevention + reporting	Nothing, or "missing data will be excluded"